

Taylor polynomial 不考

~~A1.~~ [2 marks] Let $P_2(x) = -2 + 5(x - 3) + 3(x - 3)^2$ be the degree-2 Taylor polynomial of $y = f(x)$ centred at $x = 3$. Find the equation of the tangent line to $y = f(x)$ at the point $(3, f(3))$.

~~A2.~~ [2 marks] Let $f(x)$ be a function with the following properties:

$$f(4) = 3, \quad f'(4) = -1, \quad f''(4) = 6, \quad f'''(4) = -5$$

Write the degree-3 Taylor polynomial of $f(x)$ centred at $x = 4$.

A1. [1 mark] What is the largest finite value of b for which the following integral is improper?

$$\int_b^1 \left(\frac{2 + \sin(x)}{(x+1)^2} \right) dx$$

$\rightarrow x \in \mathbb{R}$ domain: $x \neq -1$

$\int_{-\infty}^1 \rightarrow$ 可以 $\int_{-2}^1 \rightarrow$ 也可以 $\therefore$ 只能 $\int_{-1}^1 \frac{2 + \sin(x)}{(x+1)^2} dx$

$b = -1$

A7. [2 marks] Consider the differential equation

$$\frac{dy}{dx} = y^2 - y - 6 = (y-3)(y+2)$$

The solutions of this differential equation are functions of x . Sketch the graph(s) of the equilibrium solution(s) on the axes below. DO NOT SKETCH ANY OTHER SOLUTIONS.



$$\frac{dy}{dx} = 0 \Rightarrow (y-3)(y+2) = 0$$

$\therefore y = 3$ or $y = -2$

eq solutions

A8. [3 marks] Let $P(t)$ be the population of bacteria in a petri dish after t seconds. The rate of change of this population with respect to time is proportional to the number of bacteria in the petri dish. Write an example of a differential equation involving P that agrees with the previous sentence.

if proportional to time
and inversely proportional
to number
 $\frac{dP}{dt} = k \cdot \frac{t}{P}$

growth function

inversely proportional

$$\frac{dP}{dt} = k \cdot \frac{1}{P}$$

$$\frac{dP}{dt} = k \cdot P$$

$k > 0$ growth

$k < 0$ decay

B1.(a) [1 marks] Does the following integral converge or diverge?

Circle the correct answer:

$$\int_1^{\infty} \frac{e^{-x}}{(x+3)^2} dx$$

converge

diverge

B1.(b) [2 marks] Write an integral of the form $\int_1^{\infty} \frac{1}{x^p} dx$ that can be compared to $\int_1^{\infty} \frac{e^{-x}}{(x+3)^2} dx$ in order to explain why your solution to part (a) is correct.

$\rightarrow$ explain: By comparison
 $\int_1^{\infty} \frac{1}{x^2} dx$ larger integral converges, smaller integral also converges.

B1.(c) [3 marks] Explain in 1-5 sentences how your answer from (b) can be compared to $\int_1^{\infty} \frac{e^{-x}}{(x+3)^2} dx$ to conclude your answer from (a).

We choose $\int_1^{\infty} \frac{1}{x^2} dx$ because

$$\frac{e^{-x}}{(x+3)^2} = \frac{1}{e^x(x+3)^2} \text{ as } x \text{ increases, } \frac{1}{e^x(x+3)^2} < \frac{1}{(x+3)^2}$$

as we further remove constant 3, we can

conclude $\frac{1}{(x+3)^2} < \frac{1}{x^2} \therefore \int_1^{\infty} \frac{1}{x^2} dx$ converges as $p > 1$
 $\therefore$ By comparison, $\int_1^{\infty} \frac{e^{-x}}{(x+3)^2} dx$ converges.

B2.(a) [3 marks] Find the general solution of the following differential equation. Show your work.

$$\frac{dy}{dx} = 2y - xy$$

$$\frac{dy}{dx} = y(2-x)$$

$$\int \frac{1}{y} \cdot dy = \int (2-x) dx$$

$$\ln|y| = 2x - \frac{1}{2}x^2 + C$$

$$|y| = e^{2x - \frac{1}{2}x^2 + C} = e^{2x - \frac{1}{2}x^2} \cdot e^C = C \cdot e^{2x - \frac{1}{2}x^2}$$

$$\therefore y = C \cdot e^{2x - \frac{1}{2}x^2} = \text{general solution}$$

B2.(b) [3 marks] A particle's velocity $v(t)$ at t seconds is measured in millimetres per second, where

mm/s

$$\frac{dv}{dt} = 2v - tv$$

For this question $t \geq 0$. The initial velocity of the particle is 6 mm/min. Use your solution from part (a) to find the velocity function for this particle. Show your work. You can refer to your answer from the previous question.

$$\therefore V = C \cdot e^{2t - \frac{1}{2}t^2}$$

$$\therefore V = 0.1 \cdot e^{2t - \frac{1}{2}t^2}$$

6 mm/min $\xrightarrow{\div 60}$? mm/s 代入

$$t=0 \quad V(0) = \frac{6}{60} \text{ mm/s} = 0.1 \text{ mm/s}$$

$$0.1 = C \cdot e^{0-0} \quad \therefore C = 0.1$$

B3. Consider the differential equation

$$\frac{dy}{dx} = \frac{x}{2y}$$

where $y(2) = 6$.

In this question, we will approximate a solution for y .

B3.(a) [1 mark] Calculate $y'(2)$

$$\frac{dy}{dx} = \frac{x}{2y} \Big|_{\substack{x=2 \\ y=6}} = \frac{2}{2 \cdot 6} = \frac{1}{6}$$

$$\begin{array}{l} | \substack{x=2 \\ y=6} \\ = \frac{1}{2y} + x \cdot \left(-\frac{1}{2y^2}\right) \frac{dy}{dx} \\ \underline{y(2)=6} \quad \underline{y'(2)=\frac{1}{6}} \end{array}$$

B3.(b) [2 marks] Calculate $y''(2)$

$$\begin{array}{l} \frac{dy}{dx} = \frac{x}{2y} \quad \frac{d^2y}{dx^2} = 1 \cdot \frac{1}{2}y^{-1} + x \cdot \left(\frac{1}{2}y^{-1}\right)' \\ = x \cdot \frac{1}{2}y^{-2} = 1 \cdot \frac{1}{2}y^{-1} + x \cdot \frac{1}{2} \cdot (-1) \cdot y^{-2} \cdot \frac{dy}{dx} \\ = \frac{1}{2 \cdot 6} + 2 \cdot \left(-\frac{1}{2 \cdot 6^2}\right) \cdot \frac{1}{6} \\ = \frac{1}{12} - \frac{1}{6^3} \end{array}$$

~~B3.(c)~~ [3 marks] Use your answers to parts (a) and (b) of this question to calculate a Taylor polynomial of degree 2 for the solution of the differential equation y , centered at $x = 2$.

~~B3.(d)~~ [1 mark] Use your approximation to write an expression that is equal to an estimated value of $y(2.25)$. You do not have to simplify this expression.